A QUASI-AFFINE SCHEME LOCALLY OF FINITE TYPE OVER A BASE ADMITS AN IMMERSION INTO AN AFFINE SPACE OVER THAT BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 9372-9378 and proof span 9380-9410 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. This card was previously skipped because the shared gate substring-matched my chapter name against the published items of more-morphisms.tex; the Lead repaired the gate to exact file-name equality and the candidate was re-gated (RESULT=OK) before being opened.

Editorial note. The proof names the coordinates of the affine space as the chart functions together with the chosen generators; the author's notation is preserved.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A quasi-affine scheme is quasi-compact and identifies with an open subscheme of the spectrum of its ring of global functions, and the open sets where a global function is invertible and the corresponding localized ring is the ring of sections form a basis when they are affine. Cited at proof lines 9381-9386. Supplies the finite cover by standard affine opens used to build the immersion.
- The same-chapter characterization of locally finite type morphisms by finite type ring maps on affine charts. Cited at proof lines 9389-9391. Makes each coordinate ring of a chart a finite type algebra over the coordinate ring of an affine open of the base.
- The construction of a morphism into an affine space over the base from global sections of the structure sheaf and the identification of preimages of standard opens. Used at proof lines 9394-9405. Produces the candidate morphism and identifies the preimage of the standard open cut out by the j-th coordinate with the j-th chart.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01P9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2222–2222. Same-chapter reference used in the retained author text at source line(s) 2266, 9383, 9386. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\pi : X \to S$ be a morphism of schemes. Assume that X is quasi-affine and that π is locally of finite type. Then there exist $n \geq 0$ and an immersion $i : X \to \mathbf{A}^n_S$ over S.*

Proof. Let $A = \Gamma(X, \mathcal{O}_X)$. By assumption X is quasi-compact and is identified with an open subscheme of $\mathrm{Spec}(A)$, see Properties, Lemma 01P9. Moreover, the set of opens X_f, for those $f \in A$ such that X_f is affine, forms a basis for the topology of X, see the proof of Properties, Lemma 01P9. Hence we can find a finite number of $f_j \in A$, $j = 1, \ldots, m$ such that $X = \bigcup X_{f_j}$, and such that $\pi(X_{f_j}) \subset V_j$ for some affine open $V_j \subset S$. By Lemma 01T2 the ring maps $\mathcal{O}(V_j) \to \mathcal{O}(X_{f_j}) = A_{f_j}$ are of finite type. Thus we may choose $a_1, \ldots, a_N \in A$ such that the elements $a_1, \ldots, a_N, 1/f_j$ generate A_{f_j} over $\mathcal{O}(V_j)$ for each j. Take $n = m + N$ and let

$$i : X \longrightarrow \mathbf{A}^n_S$$

be the morphism given by the global sections $f_1, \ldots, f_m, a_1, \ldots, a_N$ of the structure sheaf of X. Let $D(x_j) \subset \mathbf{A}^n_S$ be the open subscheme where the jth coordinate function is nonzero. Then for $1 \leq j \leq m$ we have $i^{-1}(D(x_j)) = X_{f_j}$ and the induced morphism $X_{f_j} \to D(x_j)$ factors through the affine open $\mathrm{Spec}(\mathcal{O}(V_j)[x_1, \ldots, x_n, 1/x_j])$ of $D(x_j)$. Since the ring map $\mathcal{O}(V_j)[x_1, \ldots, x_n, 1/x_j] \to A_{f_j}$ is surjective by construction we conclude that $i^{-1}(D(x_j)) \to D(x_j)$ is an immersion as desired. $\square$